

Ethan Haines

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EDUCATION

University of Florida

Bachelor of Science in Computer Science, cum laude

Gainesville, FL

May 2026

State College of Florida

Associate's in Liberal Arts

North Port, FL

Fall 2022 – August 2023

EXPERIENCE

Software Automation Engineer

American Sanctuary · Full-time

Jul 2026 – Present

Bartow, Florida, United States · Remote

- Lead design of an evidence-backed B2B due-diligence platform that automates civil-engineering research across zoning, GIS, environmental constraints, stormwater permits, and regulatory requirements.
- Translated a 33-item engineering due-diligence workflow into a traceable domain model and requirements system spanning authoritative data acquisition, deterministic calculations, AI-assisted extraction, human review, and safe abstention.
- Utilized an agentic software-development workflow using Hermes Agent and Codex, with scoped delegation, fresh-context code review, verification gates, Git-based change control, and manual review of code.
- Researching structured government data integrations including ArcGIS REST/GIS services, FEMA, USDA/SSURGO, SWFWMD, FDOT, municipal codes, and county parcel systems, prioritizing deterministic APIs over model-driven browser automation.

PROJECTS

NEST (Nearest Extant Similarity Tool) | *Python, PyTorch, FAISS, OpenCV*

January 2026 – June 2026

- Engineered a computer vision similarity engine for fossil pollen analysis using DINOv3 embeddings and FAISS cosine similarity search to retrieve nearby extant grain matches.
- Collaborated with the Curator of Paleobotany and the Assistant Curator of Artificial Intelligence at the Florida Museum of Natural History to align NEST with museum research workflows.
- Designed an OpenCV slide-cropping and annotation workflow with persisted markers, species normalization, and metadata generation for downstream search.
- Ran embedding extraction with flip/rotation augmentation, batch inference, and GPU/CPU fallback to improve robustness across grain orientations.
- Processed and indexed approximately 3000 grain images across 16+ species, then exported interactive t-SNE views and web-ready graph JSON (nodes/edges/manifest) for research demos.

ALAN (Academic LLM Analysis Network) | *Python, Bittensor, LLM Evaluation*

December 2025 – May 2026

- Prototyped a Bittensor subnet for academic peer review quality evaluation with a custom miner/validator protocol for structured review submissions.
- Built an OpenReview ingestion and preprocessing pipeline that normalizes reviews, generates paper metadata, and splits per-miner datasets for local subnet testing.
- Authored an LLM rubric scorer (6 weighted criteria) with JSON validation, caching, and confidence scaling for consistent review scoring.
- Orchestrated grouped-by-paper ranking and normalized reward assignment (exponential rank decay) to simulate validator scoring across miner responses.

TECHNICAL SKILLS

Languages: Python, Rust, JavaScript, SQL, C++, Java, C, HTML/CSS, Lua

Frameworks/Libraries: PyTorch, Transformers, FAISS, OpenCV, NumPy, scikit-learn, Matplotlib, React, Node.js, Express, Requests

Tools/Platforms: Git, Linux, Unix, Microsoft Azure, Postman, Jira, GNU Debugger, Bittensor, OpenReview, OpenAI API, Kalshi API

Core Competencies: REST APIs, Data Pipelines, Async Programming, Distributed Systems, Machine Learning, Computer Vision, LLM Evaluation, Similarity Search, Risk Management, Agile